Problem set 1

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April 2026

Question 1

a) The individual treatment effect is $Y_i(1) - Y_i(0)$. Since only one of $Y_i(1)$ and $Y_i(0)$ can be observed the individual treatment effect can never be observed. It can be estimated by comparing the average outcome of the treated and those who are not treated, but never observed.

For the causal effect to be well defined we need the SUTVA condtions.
1. $\forall i, D_i \in \{0, 1\}$
2. $\forall i, Y_i(D_1, ..., D_N) = Y_i(D_i)$

The first condition specifies what treatment means. If D coud take on more values than 0 and 1 then $Y_i(1) - Y_i(0)$ cannot be described as the treatment effect anymore. The second condition makes sure that $Y_i(D_i = d)$ doesn't mean different things depending on how others were assigned the treatment.

2. The ATT is defined as:

$$\begin{aligned}\tau &= E[Y_i(1) - Y_i(0)|D_i = 1] = \\ &= (E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0]) + (E[Y_i(0)|D_i = 0] - E[Y_i(0)|D_i = 1])\end{aligned}$$

If the last parenthesis is zero then τ is identified. I.e. the expected untreated potential outcome does not differ between the treatment and control group. '

3. The conditional ATT is defined as:

$$\begin{aligned}\tau(x) &= E[Y_i(1) - Y_i(0)|D_i = 1, X_i = x] = \\ &= (E[Y_i(1)|D_i = 1, X_i = x] - E[Y_i(0)|D_i = 0, X_i = x])+ \\ &(E[Y_i(0)|D_i = 0, X_i = x] - E[Y_i(0)|D_i = 1, X_i = x])\end{aligned}$$

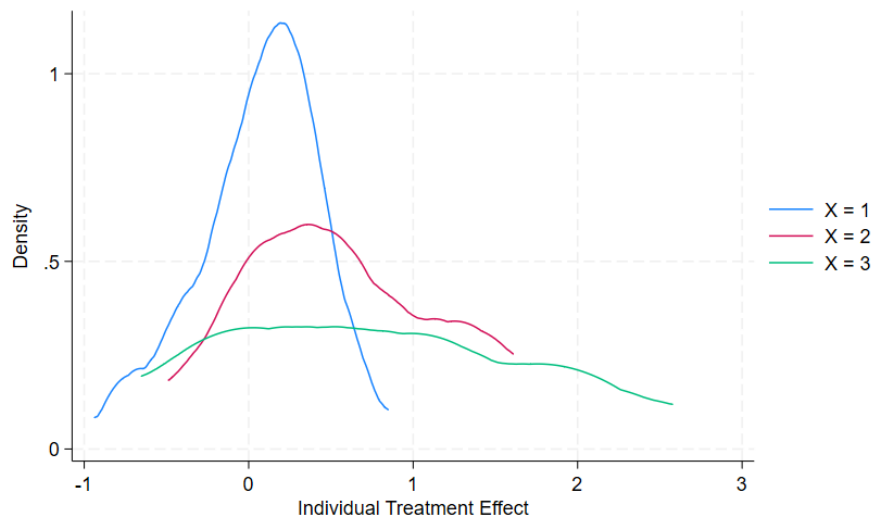
With the same argument as above we know that this is defined is the last parenthesis is zero. Also we need to have that the propensity score, $e(x)$, is nonzero and not one. Otherwise the first or the last term in the first parenthesis is nonobservable.

4. Using the expression above we get:

$$\tau(1) = 2 - (5+3)/2 = -2$$
$$\tau(2) = 4 - 5 = -1$$

and $\tau(3)$ is not identified.

6. The densities becomes:



7.

Measure	Estimated value	True value
ATE	0.569	0.37
ATT	0.569	0.551
$\tau(1)$	0.011	0.065
$\tau(2)$	0.54	0.54
$\tau(3)$	Not identifiable	0.789

The estimated ATE differs widely from the true one since the MAR assumption is not satisfied. Hence ATE is not identifiable.

Question 2

1. Looking at 1.2 above we know that the last parenthesis here is not necessarily zero. A reason, among many, for this could be that men have a lower potential untreated outcome, and also a higher probability of being treated.

2. Now we estimate the CATT (if $e(x) \in (0,1)$) for both men and women as before.

$$\begin{aligned}\tau(x) &= E[Y_i(1) - Y_i(0)|D_i = 1, X_i = x] = \\ &= (E[Y_i(1)|D_i = 1, X_i = x] - E[Y_i(0)|D_i = 0, X_i = x]) + \\ &(E[Y_i(0)|D_i = 0, X_i = x] - E[Y_i(0)|D_i = 1, X_i = x])\end{aligned}$$

We know that conditional on sex, the last parenthesis is zero. We have:

$$\tau(x) = (E[Y_i(1)|D_i = 1, X_i = x] - E[Y_i(0)|D_i = 0, X_i = x])$$

Which is observable in our sample. By the law of iterated expectations, taking expectations over X we have:

$$E[\tau(X)] = E[Y_i(1) - Y_i(0)|D_i = 1] = \tau$$

With the given values:

$$\begin{aligned}\tau(0) &= 1 \\ \tau(1) &= -1 \\ \tau &= E[\tau(X)] = 0.7 \times 1 + 0.3 \times -1 = 0.4\end{aligned}$$

3. With the same argument as before. If the propensity score is 1 or 0 for some group then averages within that group cannot be compared. Hence the CATT is not identified. The ATT is still identifieable though even if a specific groups CATT is not identifiable (I guess).

4. $N = 5, n_1 = 3$. 3 treatment units can be chosen out of 5 in $C_3^5 = 10$ different ways.

5. We have 3 men and 2 women. We can pick two men out of 3 in $C_2^3 = 3$ different ways. For each of these combinations we can pick one woman out of 2 in 2 different ways. In total we have $2 \times 3 = 6$ combinations. $p_i = e(X_i = 0)2/3$ for men, $p_i = e(X_i = 1) = 1/2$ for women. Individual probabilities and propensity scores are equal. This is since assignment probabilities are individualistic and unconfounded.

6. They are exactly equal $\bar{\tau}_{OLS} = \bar{\tau} = 0.483$. This is since:

$$\begin{aligned}
\tau_{OLS} &= \frac{\hat{Cov}(D_i, Y_i)}{\hat{Var}(D_i)} = \frac{1}{n^{-1}\sum_i (D_i - \bar{D})^2} n^{-1} \sum_i (Y_i - \bar{Y})(D_i - \bar{D}) = \\
&= \frac{1}{n^{-1}[n_0 \bar{D}^2 + n_1(1-\bar{D})^2]} n^{-1} \sum_i Y_i D_i - \bar{Y} D_i - \bar{D} Y_i + \bar{D}\bar{Y} = \\
&= \frac{1}{(1-\bar{D})\bar{D}^2 + \bar{D}(1-\bar{D})^2} (\frac{n_1}{n}\bar{Y}_1 - \bar{D}\bar{Y}) = \\
&= \frac{1}{\bar{D}(1-\bar{D})} (\bar{D}\bar{Y}_1 - [\frac{n_0}{n}\bar{Y}_0 + \frac{n_1}{n}\bar{Y}_1]\bar{D}) = \\
&= \frac{1}{\bar{D}(1-\bar{D})} (\bar{D}\bar{Y}_1 - [(1-\bar{D})\bar{Y}_0 + \bar{D}\bar{Y}_1]\bar{D}) = \\
&= \frac{\bar{D}(1-\bar{D})}{\bar{D}(1-\bar{D})} (\bar{Y}_1 - \bar{Y}_0) = \bar{Y}_1 - \bar{Y}_0
\end{aligned}$$

7. We get $\bar{\tau}(0) = 0.963, \bar{\tau}(1) = -0.605$. From a regression of Y_i on a fully satureted model in D_i and X_i we get:

	(1) Y
X	-0.447
	(0.048)
D	0.963
	(0.000)
X*D	-1.573
	(0.000)
_cons	0.0216
	(0.867)
N	200

p-values in parentheses

Which show the exact same estimates.

Question 3

1. We apply a similar logic as earlier. If n_1 units are treated then the number of treatments where unit 1 is treated is $C^4_{n_1-1}$. Total number of treatments are $C^5_{n_1}$. Probability of first unit being treated becomes:

$$C^4_{n_1-1}/C^5_{n_1} = 1/5, 2/5, 2/5, 4/5$$

For $n_1 = 1, 2, 3, 4$ respectively. The probability of the first two units being treated is:

$$C^3_{n_1-2}/C^5_{n_1} = 0, 1/10, 3/10, 3/5$$

For $n_1 = 1, 2, 3, 4$ respectively.

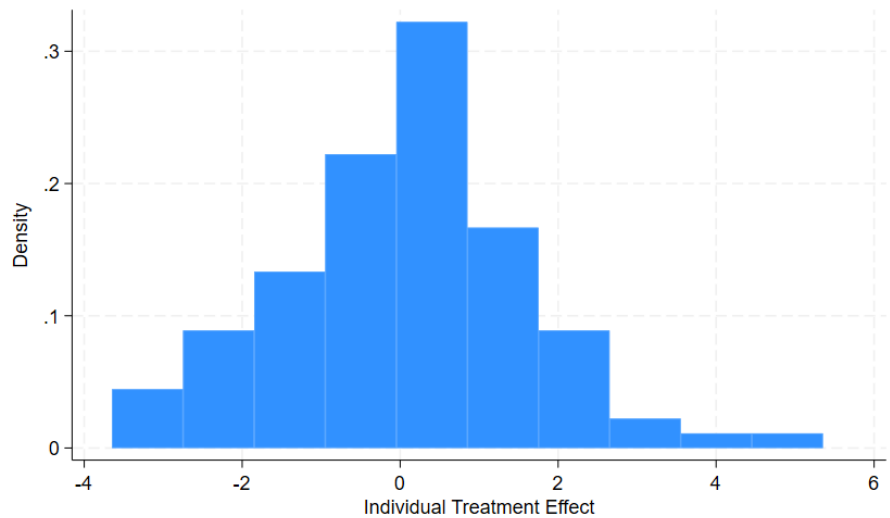
2. With $p = \mathbf{P}(D_i = 1)$

$$\tau = N^{-1}\sum_i Y_i(1) - Y_i(0) = N^{-1}Y_i(1) = N^{-1}[n_1\frac{1}{n_1}\sum_i Y_i(1) - n_0\frac{1}{n_0}\sum_i Y_i(0)] =$$
$$\frac{1}{n_1}p\sum_i Y_i(1) - \frac{1}{n_0}(1-p)\sum_i Y_i(0) = \frac{1}{n_1}\sum_{i:D_i=1} Y_i - \frac{1}{n_0}\sum_{i:D_i=0} Y_i$$

if $p\sum_i Y_i(1) = \sum_{i:D_i=1} Y_i, (1-p)\sum_i Y_i(0) = \sum_{i:D_i=0} Y_i$ which is true in expectation if the experiment is completely randomized??

3. $\bar{\tau}$ is equal to the last term above. Hence it is unbiased.

4. The distribution of individual treatment effects become:



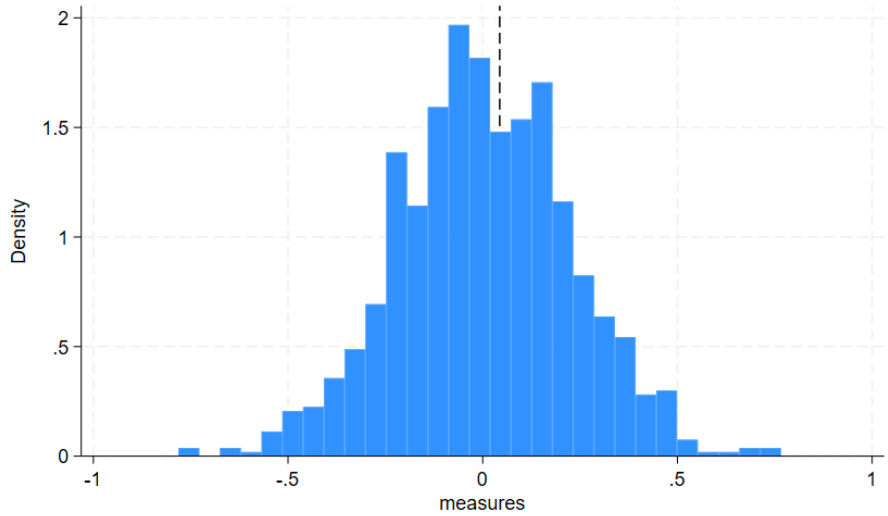
Running a regression of Y_i on D_i yields.

	(1) Y
D	0.0436
	(0.845)
Constant	-0.100
	(0.524)
N	100

p-values in parentheses

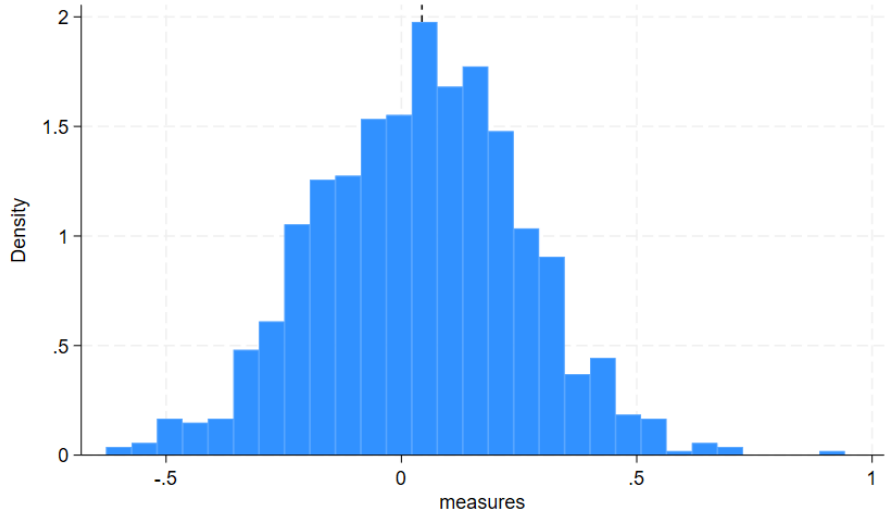
I.e no statistically significant effect, as it should be.

5. Itterating over 1000 different assignments we get the distribution of OLS estimates:



42% of estimates are larger than 0.0436.

6. Iterating over 1000 bootstraps yields distribution of OLS-estimates:



58.6% of estimates are larger than zero.